

# CBSE Class 10 Mathematics — Constructions

Time Allowed: 3 Hours

Maximum Marks: 80

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## General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

## Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. To draw a pair of tangents to a circle which are inclined to each other at an angle of  $60^\circ$ , it is required to draw tangents at the endpoints of those two radii of the circle, the angle between which is:  
(A) (A)  $60^\circ$   
(B) (B)  $90^\circ$   
(C) (C)  $120^\circ$   
(D) (D)  $180^\circ$
2. If a point  $P$  is 10 cm away from the centre of a circle of radius 6 cm, the length of the tangent drawn from  $P$  to the circle is:  
(A) (A) 7 cm  
(B) (B) 8 cm  
(C) (C) 9 cm  
(D) (D) 10 cm
3. How many tangents can be drawn from an external point to a circle?  
(A) (A) 1  
(B) (B) 2  
(C) (C) 0  
(D) (D) Infinite
4. To divide a line segment  $AB$  in the ratio 3 : 4, we draw a ray  $AX$  such that  $\angle BAX$  is acute. Then we mark points on  $AX$  at equal distances. The minimum number of points to be marked is:  
(A) (A) 3  
(B) (B) 4  
(C) (C) 7

(D) (D) 12

5. The angle between a tangent to a circle and the radius passing through the point of contact is:
- (A) (A)  $45^\circ$
  - (B) (B)  $60^\circ$
  - (C) (C)  $90^\circ$
  - (D) (D)  $180^\circ$
6. Assertion (A): The length of the tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
  - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
  - (C) (C) Assertion (A) is true but Reason (R) is false
  - (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): To construct a pair of tangents to a circle from an external point, the angle between the tangents is supplementary to the angle subtended by the line segment joining the points of contact at the center. Reason (R): The sum of opposite angles in a cyclic quadrilateral is  $180^\circ$ .
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
  - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
  - (C) (C) Assertion (A) is true but Reason (R) is false
  - (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): It is impossible to construct a tangent to a circle from a point lying inside the circle. Reason (R): A tangent touches the circle at only one point.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
  - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
  - (C) (C) Assertion (A) is true but Reason (R) is false
  - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The number of tangents that can be drawn from a point on the circle to the circle is exactly two. Reason (R): A point on the circle lies on the circumference.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): When constructing tangents to a circle from an external point  $P$ , we bisect the line segment  $OP$  where  $O$  is the center. Reason (R): The perpendicular bisector of  $OP$  helps in finding the midpoint which acts as the center of a circle passing through  $O$  and  $P$ .

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

### Section C: Short Answer I (3 marks each)

1. Draw a circle of radius 3 cm. Take two points  $P$  and  $Q$  on one of its extended diameter each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points  $P$  and  $Q$ .
2. Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.
3. Draw a line segment  $AB$  of length 8 cm. Taking  $A$  as centre, draw a circle of radius 4 cm and taking  $B$  as centre, draw another circle of radius 3 cm. Construct tangents to each circle from the centre of the other circle.
4. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of  $60^\circ$ .
5. Draw a circle of radius 5 cm. Mark a point  $P$  at a distance of 8 cm from the centre. Construct the two tangents from  $P$  to the circle.

### Section D: Long Answer (4-5 marks each)

1. Draw a circle of radius 3 cm. Take two points  $P$  and  $Q$  on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points  $P$  and  $Q$ .
2. Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.

3. Draw a line segment  $AB$  of length 8 cm. Taking  $A$  as center, draw a circle of radius 4 cm and taking  $B$  as center, draw another circle of radius 3 cm. Construct tangents to each circle from the center of the other circle.
4. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of  $60^\circ$ .
5. Construct a triangle with sides 5 cm, 6 cm and 7 cm. Then construct a circle touching the side of 6 cm at its midpoint and passing through the other vertex.
6. Draw a circle of radius 3 cm. Take two points  $P$  and  $Q$  on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points  $P$  and  $Q$ .
7. Construct a triangle  $ABC$  with side  $BC = 6$  cm,  $AB = 5$  cm and  $\angle ABC = 60^\circ$ . Then construct a triangle whose sides are  $\frac{3}{4}$  of the corresponding sides of the triangle  $ABC$ . Finally, construct a tangent to the circumcircle of triangle  $ABC$  at vertex  $A$ .
8. Draw a pair of tangents to a circle of radius 4 cm, which are inclined to each other at an angle of  $60^\circ$ . Justify the construction.
9. Draw two concentric circles of radii 3 cm and 5 cm. Construct a tangent to the smaller circle from a point on the larger circle and measure its length. Verify the measurement by calculation.